Spin for Abilities – Core Combat (M1, Block, Dash)

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Overview

✅ **Every player has M1, Block and Dash by default.** They are never rolled, never take a skill slot, and work the same for everyone. Rolled abilities sit on top of this base. The feel is taken from The Strongest Battlegrounds (TSB): fast 4-hit M1 strings, an uppercut into an air combo, F to block, Q to dash, and ragdoll cancel to escape a knockdown.

Key	Move	What it does
M1	Ground combo	4-hit punch string; hit 4 knocks back
Hold Space + M1 (before hit 4)	Uppercut	Launches the enemy up and starts the air combo
M1 (in the air)	Air combo → Downslam	3 air hits, the last one slams them down
W + Q (enemy at a wall, after hit 4)	Wall combo	Bonus hit, enemy left on the ground
F (hold)	Block	Front damage reduced; can be guard broken
Q / W + Q	Front dash	Long dash forward
A + Q / D + Q	Side dash	Short sidestep with i-frames
S + Q	Back dash	Hop away
Q (while ragdolled)	Ragdoll cancel	Recover early, on its own cooldown

- All numbers below are **starting values** and live in the config module `CombatConfig.luau`, so balance changes never touch combat code.
- Rolled abilities can't be used while stunned or ragdolled, but can cancel M1 end lag (see Combo rules).

M1 ground combo

M1 is a 4-hit string. Every hit stuns the target just long enough for the next M1 to land, so a player caught in the string can't attack back. If the attacker stops clicking, the stun runs out and the target is free.

Hit	Damage	Hitstun on target	Window for next M1	What it looks like
1	3	0.45 s	0.15–0.55 s after hit	Quick jab, small white impact puff
2	3	0.45 s	0.15–0.55 s after hit	Cross punch, impact puff + tiny shake
3	3	0.45 s	0.15–0.55 s after hit	Hook, bigger puff, target staggers
4 (finisher)	5	Knockback + ragdoll 1.2 s	String ends	Heavy punch, shock ring, camera shake, target flies ~18 studs

The stun rule

Hitstun must always be **longer than the time between M1s**. With an M1 every 0.35 s and 0.45 s of hitstun, the next hit always lands before the target recovers. That 0.1 s gap is the whole "can't attack back" feeling, so it is one config value we tune, not something hidden in animations.

- While stunned the target **can't** M1, use skills, dash or jump. They **can** hold Block only if it was already held before the first hit landed (a "true block" read), which stops M1 spam from being unbeatable.
- Walk speed drops to 4 while stunned (normal 16), so they can't walk out of range.
- The attacker gets a small forward step (2 studs) on hits 1–3 so the string doesn't drift out of range.

Combo rules

- **Combo reset:** no M1 within 0.55 s → the string goes back to hit 1.
- **End lag:** after hit 4 the attacker can't M1 for 1.5 s. Dash and rolled skills *can* be used during end lag, which is how players extend combos (M1 ×4 → skill).

- **Miss:** a whiffed M1 still advances the string but adds 0.2 s recovery, so spamming at nothing is punished.
- **Damage scaling:** each hit after the 8th in one combo deals 10% less (min 50%), so infinite combos stay unrewarding.
- **Stun immunity:** after being ragdolled, a player can't be stunned again for 0.6 s after standing up, so they can't be locked forever.

Air combo

Hold Space before the 4th M1 to swap the finisher for an Uppercut. Both players go airborne, the attacker lands 3 air M1s, and the last one is a Downslam. This is the TSB pattern: uppercut has no pushback, downslam goes through block and leaves the enemy on the ground.

Step	Input	Damage	Effect	What it looks like
Uppercut	Hold Space, then M1 as hit 4	4	Target launched ~14 studs straight up, attacker jumps after them	Rising fist, upward wind streak, dust ring on the ground
Air hit 1	M1 in air	3	0.45 s hitstun, both float	Small impact puff
Air hit 2	M1 in air	3	0.45 s hitstun, both float	Impact puff
Downslam	M1 in air (3rd air hit)	6	Target slammed to the ground, knocked down 1.0 s; **ignores block**	Two-handed hammer blow, dust crater + shock ring, heavy shake

- **Hang time:** during the air combo both players have gravity set to 20% and vertical velocity zeroed on each hit, so they float in place. If the attacker doesn't M1 within 0.55 s, gravity returns and both fall.
- **Early downslam:** M1 while holding S in the air does the downslam straight away, for a shorter but safer combo.
- **Air combo cooldown:** a target hit by a Downslam can't be uppercut again for 4 s, so air loops can't repeat.

- **Skills in the air:** rolled abilities tagged `AirUsable` (for example Lightning Step, Meteor Drop) can follow an uppercut instead of air M1s.

Block (F)

Hold F to cut damage from the front by 70% and take no hitstun from M1s. Block has a guard meter: 6 blocked hits break it and stun you for 1.5 s. Block stops M1 spam, and guard break stops players from turtling forever.

Rule	Value	Notes
Damage taken while blocking	30%	Only attacks from the front 120° cone
Hits from behind	Full damage + stun	Block does nothing from behind, so dashing around a blocker works
Walk speed while blocking	8 (normal 16)	Can't jump, dash or M1 while held
Guard meter	6 points	Each blocked M1 = 1, blocked skill = 2, heavy skill = 3
Guard meter regen	1 point per 1 s not blocking	Starts 1.5 s after the last blocked hit
Guard break stun	1.5 s	Block disabled for 3 s after
Block startup	0.1 s	Prevents "panic block" mid-string; must be held before the hit

Always goes through block: Downslam, the Haymaker skill, grabs (Grab & Throw), and any skill tagged `GuardBreak`.

What it looks like: forearms crossed in front, a faint blue shield shimmer on each blocked hit with a "tink" sound. The shimmer cracks at 2 points left. Guard break = glass-shatter effect, the player staggers back holding their arms.

Dash (Q)

Q dashes in the direction you're holding. Front dash covers ground, side dash dodges with i-frames, back dash escapes. Side and back dash share one cooldown; front dash has its own, so you can engage and still dodge.

Dash	Input	Distance	I-frames	Cooldown	Special
Front	Q or W + Q	30 studs over 0.35 s	None	2.5 s	Can be cancelled into a side dash; hits a wall-pinned enemy (wall combo)
Side	A + Q / D + Q	14 studs over 0.25 s	0.2 s	2.0 s (shared)	Curves slightly toward the locked target
Back	S + Q	16 studs over 0.3 s	0.1 s	2.0 s (shared)	Small hop

Wall combo

After the 4th M1, if the enemy hits a wall within 0.6 s, a front dash into them does a shoulder-check: **+12% bonus damage** and leaves them knocked down 1.0 s. The server checks the target's last knockback with a raycast toward their velocity (3 studs) to confirm the wall.

Dash cancel

Front dash into side dash (press A/D + Q mid-dash) changes direction instantly. It uses the side dash cooldown. This is the main movement tech that good players learn.

What it looks like: front dash = speed lines + dust kicked behind; side dash = afterimage of the player for 0.15 s; back dash = small dust puff. No dash while stunned, ragdolled or blocking.

Ragdoll and ragdoll cancel

Knockback attacks put the target into ragdoll. Pressing Q during ragdoll does a ragdoll cancel: you snap back to your feet with a short dodge, on a 20 s cooldown. It's the escape button that stops one finisher from turning into a free follow-up.

What causes ragdoll

Source	Ragdoll time
M1 hit 4 (finisher)	1.2 s
Downslam	1.0 s (knocked down, not flung)
Wall combo	1.0 s
Skills tagged `Ragdoll` (Front Kick, Wind Gust, Earth Spikes, Gravity Well…)	Set per skill

While ragdolled the player can't move, attack, block or use skills, and they still take damage, which is what makes follow-ups strong.

Ragdoll cancel rules

Rule	Value
Input	Q while ragdolled
Earliest use	0.25 s after the ragdoll starts (so the finisher still feels like it landed)
Result	Player stands up instantly and does a short back dash (10 studs)
I-frames	0.4 s
Cooldown	20 s (shown as its own icon next to the dash icon)
Dash cooldown	Not used, so you can still dash after cancelling
Can't cancel	Downslam knockdown, Time Stop, Black Sun, grabs

What it looks like: a white-blue flash around the body, the player flips back onto their feet, and a small ring pulses out. A distinct sound plays for both players so the attacker knows the cancel was used and its cooldown is now running, which becomes the window to go in again.

How we build it in Roblox

The server owns all combat: the client only sends "I pressed M1 / F / Q", and the server decides hits, damage, stun and ragdoll. The client plays animations and effects straight away so it feels instant, but it never decides who got hit.

Where the code lives

```
ReplicatedStorage/Shared
  CombatConfig.luau      -- every number in this doc (damage, stun, windows, cooldowns)
  CombatTypes.luau       -- state names, remote payload types
ServerScriptService/Combat
  CombatService.luau     -- receives inputs, validates, runs the state machine
  HitboxService.luau     -- spatial query hit detection
  RagdollService.luau    -- ragdoll on/off, ragdoll cancel
  StateService.luau      -- per-player state + timers (Stunned, Blocking, Ragdolled...)
StarterPlayerScripts/Combat
  CombatController.luau  -- reads input (ContextActionService), sends remotes
  CombatVFX.luau         -- animations, effects, camera shake, sounds
  CombatHUD.luau         -- cooldown icons, guard meter
```

Player states

Each player is in exactly one state at a time, stored on the server (and mirrored to the client as an attribute for UI). Every input is checked against the current state first.

State	Can M1	Can block	Can dash	Can use skills	Leaves when
Idle	Yes	Yes	Yes	Yes	Any action
Attacking	Next M1 only	No	Yes (cancels)	Yes (cancels end lag)	Swing ends
Blocking	No	—	No	No	F released or guard break
Stunned	No	Only if held before	No	No	Hitstun timer ends
Airborne combo	Yes (air M1)	No	No	`AirUsable` only	Downslam or 0.55 s idle

State	Can M1	Can block	Can dash	Can use skills	Leaves when
Ragdolled	No	No	Ragdoll cancel only	No	Ragdoll timer or cancel
GuardBroken	No	No	No	No	1.5 s timer

One M1, start to finish

1. **Client:** player clicks. `CombatController` checks local state + cooldown, plays the swing animation and sends `RequestM1` with a timestamp and the aim direction.
2. **Server validates:** player is alive, state allows M1, combo window is open, not faster than the minimum M1 interval (0.3 s, a rate limit against autoclickers and exploits). Bad requests are dropped silently.
3. **Hit detection:** at the animation's hit frame (0.12 s), `HitboxService` runs `workspace:GetPartBoundsInBox` with a 5×5×5 stud box, 3.5 studs in front of the attacker's HumanoidRootPart, using `OverlapParams` that exclude the attacker. Take the closest valid target.
4. **Apply:** check the target's block (facing angle with a dot product, guard meter), then apply damage, set `Stunned` with the hitstun timer, and set the target's walk speed to 4.
5. **Broadcast:** fire `CombatEvent` to all clients near the fight (within 150 studs) with `{attacker, target, hitIndex, blocked}`. Each client plays the impact effect locally.

Timers and stun

- Stun is a timestamp, not a `wait()`: `state.stunEnd = now + hitstun`. A new hit overwrites it, so stun never stacks or ends early by mistake.
- One `Heartbeat` loop in `StateService` checks timers for all players, instead of a thread per hit. This keeps the server cheap in 20-player servers.
- Network ownership of a stunned or ragdolled character's root part goes to the server, so the target's client can't move itself out.

Ragdoll

- On ragdoll: `Humanoid:ChangeState(Enum.HumanoidStateType.Physics)`, swap Motor6Ds for `BallSocketConstraint`s (built once per character on spawn, enabled or disabled, never created per hit), then apply the knockback with

`AssemblyLinearVelocity`.

- Ragdoll cancel: client sends `RequestRagdollCancel`; the server checks the 0.25 s earliest time and 20 s cooldown, re-enables the Motor6Ds, sets `GettingUp`, gives 0.4 s i-frames and applies the back dash velocity.

Remotes

Remote	Direction	Payload
`RequestM1`	Client → Server	`{t: number, dir: Vector3, holdingSpace: boolean, holdingS: boolean}`
`RequestBlock`	Client → Server	`{down: boolean}`
`RequestDash`	Client → Server	`{dir: "F"
`RequestRagdollCancel`	Client → Server	`{}`
`CombatEvent`	Server → Clients	`{kind, attacker, target, data}`

Every payload is type-checked on the server (wrong types = drop). Damage, positions and targets are never taken from the client.

Animations, VFX and build order

Animations are made in Roblox's Animation Editor (or Moon Animator) with a named `Hit` marker on the impact frame. The server uses fixed hit times from `CombatConfig`, and the client uses the marker to time effects. Keeping the two in sync is a checklist item every time an animation changes.

Animations needed (11)

Animation	Length	Hit frame
M1 1 / 2 / 3 / 4	0.3 / 0.3 / 0.3 / 0.45 s	0.12 s (hit 4: 0.2 s)
Uppercut	0.4 s	0.15 s
Air M1 1 / 2	0.3 s	0.12 s

Animation	Length	Hit frame
Downslam	0.45 s	0.25 s
Block idle, Block hit, Guard break	loop / 0.2 / 0.6 s	—
Front, side, back dash	0.35 / 0.25 / 0.3 s	—
Ragdoll cancel get-up flip	0.3 s	—

VFX rules

- Effects use the shared mesh kit from the abilities doc where possible (shock ring, swirl ribbon). Impact puffs are `ParticleEmitter`s with `:Emit()` on a pooled attachment.
- All combat VFX run on each client from `CombatEvent`. Nothing visual is created on the server.
- Camera shake only for the local player when they hit or get hit, and a setting to turn it off.
- Low quality mode (phones): one particle burst per hit, no shock rings on hits 1–3.

Build order

1. `StateService` + `CombatConfig`, with debug text over each player's head showing their state.
2. M1 ground string: hitbox, damage, hitstun. Test with 2 players that the stunned player really can't hit back.
3. Block + guard meter.
4. Dash (front, side, back) + dash cancel.
5. Ragdoll + hit 4 knockback, then ragdoll cancel.
6. Uppercut, air M1s, downslam.
7. Wall combo.
8. Animations, VFX, sounds, HUD cooldown icons.
9. Playtest with 8+ players at 150–250 ms ping (Studio network emulation) and retune the stun and window values.

Open question: should hitstun be lowered slightly for players with very high ping, or should we keep one value for everyone and accept some unfairness?

Sources

- All controls in Roblox The Strongest Battlegrounds — uppercut, downslam, F block, Q dash and wall combo reference. All timings and numbers in this doc are our own starting values.